UNIVERSITY OF RWANDA
College of Science and Technology · Department of Applied Mathematics

A Fractional-Order Compartmental Model of HIV Transmission with Social Behaviour Interventions

NDACYAYISABA Lambert

Reg. 222008909 · B.Sc. Applied Mathematics

Supervisor: Dr. MUHIRWA Jean Pierre

Final Year Project Report

Academic Year 2025–2026

University of Rwanda · Kigali, Rwanda · May 2026

Background / Motivation

Rwanda-aware public-health motivation

- HIV transmission is shaped by biology, treatment access, and social behaviour.
- Awareness, safer behaviour, testing, treatment-seeking, and adherence can change long-term outcomes.
- In Rwanda, continued prevention and adherence support remain important alongside treatment progress.

Why mathematics? A model gives a controlled way to compare intervention strategies and understand possible long-term behaviour.

Context
HIV in Rwanda

Mechanism
Social behaviour

Method
Fractional memory

Output
Simulation evidence

Problem Statement

Limitations in ordinary modelling

- Ordinary models usually depend only on the present state.
- HIV-related behaviour may depend on past awareness, stigma, testing history, and adherence.
- Social behaviour is often simplified into fixed rates.

Need addressed by this project

- Include memory using a Caputo fractional derivative.
- Represent behavioural interventions through effective rates.
- Use numerical simulation to compare scenarios clearly.

Research focus: How do fractional memory and social behaviour interventions affect simulated HIV dynamics in a SITA framework?

Literature Review and Research Gap

What existing studies show

- Classical HIV models commonly use ordinary differential equations and treatment compartments such as SICA/SITA.
- Previous studies show that social behaviour, testing, safer sexual practices, stigma, and adherence influence HIV transmission.
- Fractional-order epidemic models use Caputo derivatives to capture memory and hereditary effects.

Research gap

Existing work does not commonly combine:

- fractional HIV dynamics,
- social behaviour intervention functions,
- and an interactive simulation dashboard.

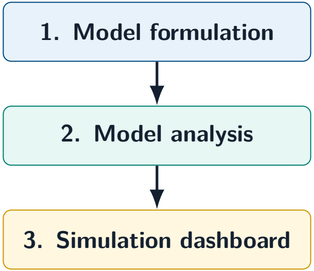
Gap addressed: a fractional-order SITA model that connects memory effects, social behaviour interventions, numerical simulation, and dashboard exploration.

Objectives

Main objective

To formulate, analyse, and computationally simulate a fractional-order SITA model of HIV transmission incorporating social behaviour interventions. **Specific objectives**

1. Formulate a fractional-order SITA HIV transmission model using the Caputo derivative and social behaviour intervention functions.
2. Analyse the main mathematical properties: positivity, boundedness, disease-free equilibrium, $\mathcal{R}_0$, and fractional-order stability.
3. Implement numerical simulations and a Python dashboard for studying memory effects, intervention scenarios, sensitivity, and epidemic trajectories.



How the Objectives Are Addressed

Objective 1

Formulation
Caputo fractional SITA system with S, I, T, A compartments and intervention functions β_{eff}, τ_{eff}, ρ_{eff}.

Objective 2

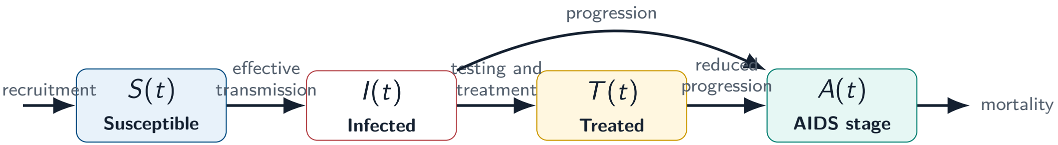
Analysis
Positivity, boundedness, disease-free equilibrium, $\mathcal{R}_0$, and fractional-order local stability are presented in the report.

Objective 3

Implementation
Python simulations and dashboard views show trajectories, memory effects, intervention scenarios, and sensitivity.

Defense point: each objective is represented in the thesis, the generated figures, and the live dashboard demonstration.

Model Diagram: SITA



State variables: $S(t)$ susceptible, $I(t)$ infected, $T(t)$ treated, $A(t)$ AIDS-stage population.

Controls: u_1 awareness, u_2 safer behaviour, u_3 testing, u_4 adherence.

Fractional-Order Model

Caputo memory form

$$
\begin{aligned}
{}^{C}D_t^{q}X(t) &= F(t, X(t)), \qquad 0 < q \leq 1 \\
{}^{C}D_t^{q}S &= \Lambda - \beta_{\text{eff}}S(I + \eta T) - \mu S, \\
{}^{C}D_t^{q}I &= \beta_{\text{eff}}S(I + \eta T) - (\tau_{\text{eff}} + \delta + \mu)I, \\
{}^{C}D_t^{q}T &= \tau_{\text{eff}}I - (\rho_{\text{eff}} + \mu)T, \\
{}^{C}D_t^{q}A &= \delta I + \rho_{\text{eff}}T - (d + \mu)A.
\end{aligned}
$$

Intervention-adjusted rates

$$
\begin{aligned}
\beta_{\text{eff}} &= \beta_0(1 - u_1)(1 - u_2) \\
\tau_{\text{eff}} &= \tau(1 + u_3) \\
\rho_{\text{eff}} &= \rho(1 - u_4)
\end{aligned}
$$

When $q = 1$, the model is ordinary. When $q < 1$, past states influence current change.

Numerical Scheme + Implementation

Numerical idea

- Time interval is divided into discrete steps.
- A predictor estimates the next state.
- A corrector updates the state using fractional-memory weights.
- One engine supports baseline, memory, intervention, and sensitivity experiments.

Inputs
parameters, initial conditions, q, controls

↓

Solver
fractional ABM-type method

↓

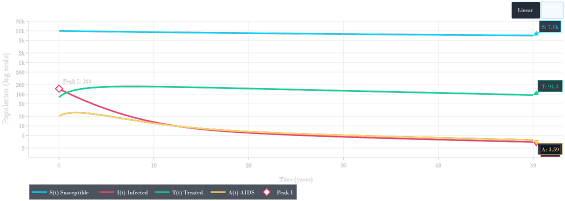
Outputs
trajectories, $\mathcal{R}_0$, scenarios, sensitivity

↓

Demo
moving graphs and scenario comparison

Presentation link: the slides explain the model; moving graphs are shown separately during the live demonstration.

Baseline Simulation Result

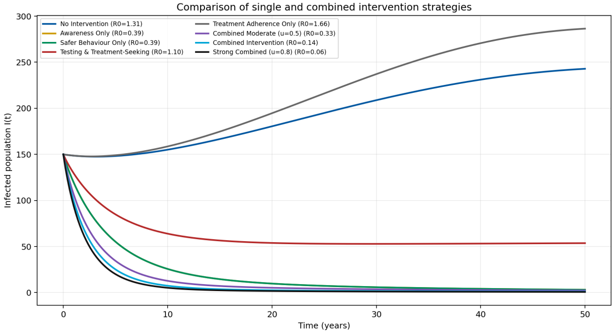


Computed baseline

- $\mathcal{R}_0 = 0.430 < 1$, below the epidemic threshold.
- Peak infected population is 150 at $t = 0$ years.
- Final values after 50 years: $I = 3.1$, $T = 94.3$, $A = 3.6$.

Interpretation: the baseline intervention-adjusted configuration moves toward disease-free behaviour.

Simulation Results: Intervention Comparison

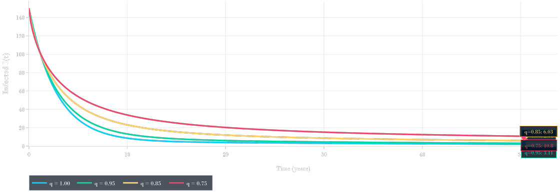


Main interpretation

- Single interventions help, but each targets only one mechanism.
- Combined intervention acts on transmission, treatment uptake, and progression.
- The combined strategy gives stronger simulated control than isolated changes.

Public-health meaning: prevention and treatment support should work together.

Simulation Results: Fractional Memory

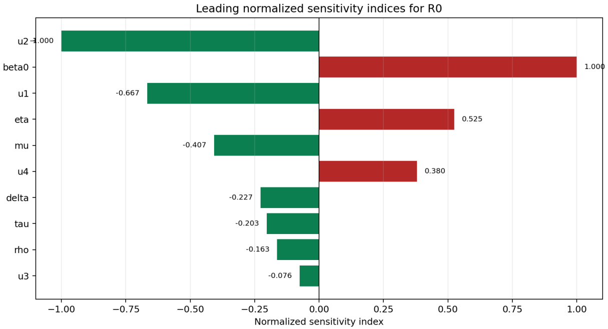


Why q matters

- $q = 1$ represents the ordinary model.
- Smaller q values strengthen the memory effect.
- Different q values produce visibly different trajectories.

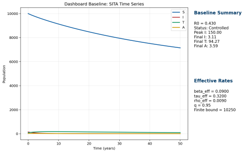
Defense point: fractional calculus is not decorative; it changes the simulated dynamics.

Dashboard and Sensitivity Evidence



Main evidence

- Transmission-related quantities dominate $\mathcal{R}_0$ sensitivity.
- Safer behaviour and awareness strongly reduce $\mathcal{R}_0$.
- Dashboard outputs support thesis discussion and live defense.

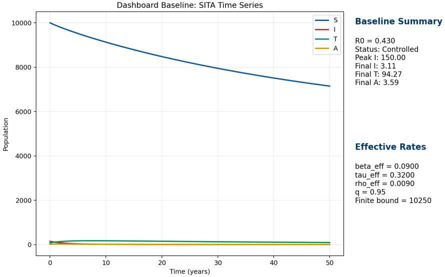


Live Dashboard Demonstration

Open during the defense

Dashboard link
`https://hiv-model.onrender.com/`

1. Run the baseline simulation.
2. Show the final state summary and interpretation.
3. Compare no intervention with combined intervention.
4. Change q to show the fractional memory effect.



After the demo: return to the next slide for conclusion and recommendations.

Conclusion + Recommendations

Conclusion

- The SITA HIV model was extended using fractional memory.
- Behavioural interventions were represented through effective rates.
- Numerical simulations support the value of combined intervention strategies.

Recommendations

- Calibrate parameters with Rwanda-specific data.
- Extend the model using age, gender, or stochastic structure.
- Compare other fractional operators.
- Improve the dashboard into an educational decision-support tool.

Final message: the work connects mathematical analysis, numerical simulation, and a working dashboard for communicating HIV intervention dynamics.